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09/09/2026 Rev 0 (always check if a newer version is available)

MirrorBallBot - Quick Start Guide

Build a ball-balancing robot that sees through a mirror

This is the Quick Start Guide, providing the essential information to build the MirrorBallBot.

For detailed explanations, theory, design philosophy, and troubleshooting, please refer to the full MirrorBallBot Reference Manual.

Robot demonstration at YouTube: https://youtu.be/ORftNl7bZG4

Wooden base version

Fully 3D printable version:

1. Table of contents

2. What is MirrorBallBot?

MirrorBallBot is a ball-balancing robot. It uses a camera, three stepper motors, and a clever mirror setup to keep a coloured ball balanced on a circular platform.

What it can do:

- Automatically balance the ball at the center of the platform
- Reject disturbances
- Move the ball to any position you choose
- Follow your finger on the touchscreen (real-time)
- Record and replay custom paths
- Execute predefined paths: square, circle, infinity, triangle, line

Two models available:

- Wooden base – my preference, heavier, dampens vibrations
- Fully 3D-printable – no woodworking required, fits a 210×210mm print bed

Project difficulty: 8/10 (using CUBOTino at 7/10 as a reference)

3. Two Instruction manuals

MirrorBallBot is documented in two complementary manuals:

Together, the two manuals serve different purposes:

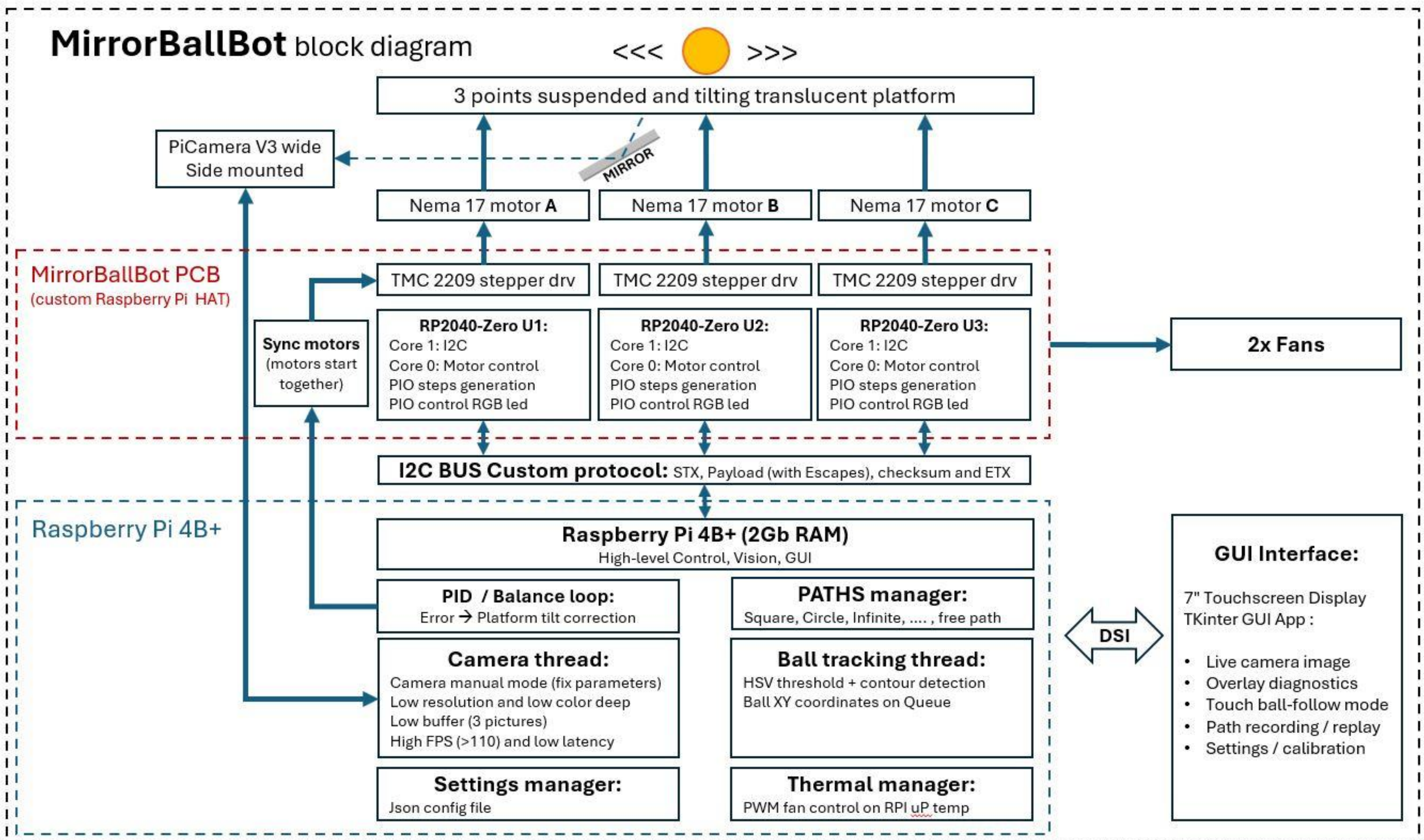
Manual	Content	Pages
Quick Start Guide (this document)	Essential information to build the robot. Concise, action-oriented, step-by-step	~ 50
Reference Manual	Detailed explanations, theory, design philosophy, troubleshooting, Q&A and much more. It also includes the assembly steps, with several pictures and detailed information.	~ 170

Which manual should you use?

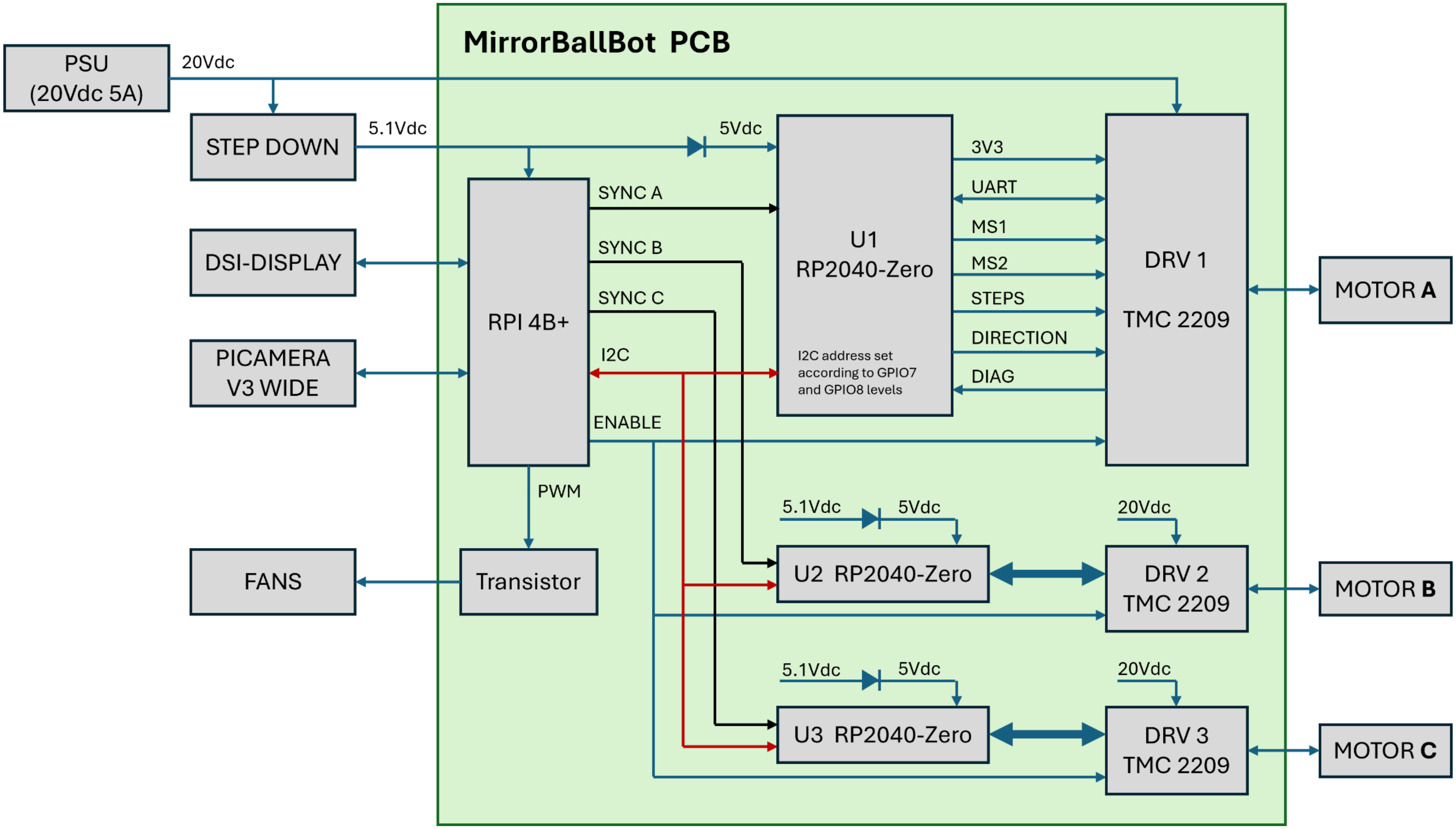
Start with the Quick Start Guide. It will get you from zero to a working robot with minimal friction. Follow the steps, and refer to the Reference Manual only when you need more detail.

Use this Reference Manual when: you want to understand *why* something works, you encounter an issue not covered here, or you're curious about the design process, the custom I2C protocol, or the PID controller theory.

4. Block diagram



Modules connections diagram



5. Safety disclaimer

Energize the robot only via a power supply having class 2 insulation from the Mains. Chargers for laptops normally have this safety feature: Check it for your own safety. Despite the robot mechanical force is limited, it must be operated only under adult supervision.

If you build and use a robot, based on this information, you accept it is on your own risk.

6. What you'll need

Estimated Time

Task	Time Estimate
3D printing (wooden version / fully 3D version)	2 days / 4 days
Wood base preparation (if applicable)	½ - 1 day
PCB soldering	½ day
Programming & testing	½ day
Assembly	½ day
Tuning & calibration	A few hours to a few days
Total	**~2-3 weeks**

Cost: Approximately **€300** in materials (spring 2026 prices).

Tools Required

- 3D printer (210×210mm bed minimum), **or** access to one
- Laser cutter (for acrylic platform and mirror) **or** access to one
- (Optional) Router or CNC for wooden base version
- Soldering iron and basic soldering skills
- Multimeter
- Screwdrivers (Phillips, flathead, Allen keys)

7. BOM

Ref	Description	Link	Cost (Euro)
1	***Raspberry Pi 4*** (2Gb or 1Gb RAM)	https://www.raspberrypi.com/products/raspberry-pi-4-model-b/	70
2	***PiCamera*** V3 wide (12Mp, not the NoIR)	https://www.raspberrypi.com/products/camera-module-3/	40
3	MicroSD 16Gb Class 10 (Speed is only relevant for the OS flashing, not for the robot operation)	Avoid cards with low ratings	10
4	Ribbon cable (15pins 30cm)	https://tinyurl.com/ssehw7dh	4.5
5	Stepper motors (42x42x48mm, 1.8deg/step, 2A, ca50Ncm, shaft Ø5.0mm, D-cut 15mm)	https://tinyurl.com/mr2fmdmp	30
6	Stepper drivers (TMC 2209 V1.3)	https://tinyurl.com/yjtc4p3f	15
7	Stepper motor controller (***RP2040-Zero***)	https://tinyurl.com/myztcukk	4.5
8	***MirrorBallBot PCB*** PCBWay	https://www.pcbway.com/	5
9	**Display Elecrow 7"** 800x600 DSI touchscreen ELECROW	https://www.elecrow.com/7-inch-800-480-dsi-display-touch-screen-with-bracket-compatible-with-raspberry-pi.html?idd=5	40
10	Brushless fan 40x10mm 24Vdc	https://tinyurl.com/2zf8nw3h	9.4
11	Fans' grids	https://tinyurl.com/yznzz6fp	5.5
12	Step-down power supply (***XL4015*** adjustable 4-35V)	https://tinyurl.com/26f39usa	2.7
13	Spheric-joints (SA5T/K Rod End Bearing, 4pcs 5mm Metric Male Thread)	https://tinyurl.com/3jdnu45t	3

Ref	Description	Link	Needed Q.ty
14	Acrylic (Plexiglass) translucent plate to laser cut Ø26cm. **3mm** thick for balancing, **6mm** thick for bouncing	https://tinyurl.com/2m9ttbz6	1
15	Acrylic (Plexiglass) mirror 2.5mm to laser cut (Self-adhesive types are easy to source)	https://tinyurl.com/bdeuabmh	1
16	Push button (12mm momentary push button)	https://tinyurl.com/36j3m7ru	2
17	Standoff screws (M2.5mm x 15mm x 6mm)	https://tinyurl.com/t3c636ns	4
18	Plywood 18mm unless you prefer the fully 3D printable version	Check at your local wood/ hardware store	1 panel (60x60 cm)
19	Adjustable feet (thread from M5 to M8)	Check at your local hardware store	3
20	Spirit level Ø60mm	https://tinyurl.com/ukpsd8r7	1
21	PETG Filament (~1.1Kg if you prefer the fully 3D printable version)	Your usual filament supplier	~0.55 Kg (~1.1 Kg)
22	Plastic's ball (suggested orange, Table football balls Ø30 ~ 35mm, 17~25 grams)	https://tinyurl.com/yc8a9b4n	1
23	Power supply 20V 5A (I used a spare power supply from a HP laptop)	https://tinyurl.com/58j5st68	1
24	Power socket (in my case one compatible with the HP power supply: Computer Adapter Cable Converter Cable 7.4x5mm Female to 5.5x2.5mm Male for Laptop)	https://tinyurl.com/bdz3cauv	1

Notes:

1. The reference to amazon.com are used as it might offer more accurate description and pictures of the products, as well as a cost reference.
2. I personally buy many parts from Aliexpress.com and other shops, also Amazon.nl, yet not necessarily a convenient choice.

Components for the PCB (all for through hole assembly)

Component	Type	Quantity
Resistors (1/4 W)	470 Ω	6
	1 KΩ	4
	10 KΩ	4
Diodes	1N5818 (or 1N5819)	4
Capacitors	100~220uF 35V electrolytic	4
	10uF 30V ceramic	7
Transistor	BC337	1
Male header 2.54mm	2 pins	5
	3 pins	1
	4 pins	3
Female headers 2.54mm	2pins	3
	7pins	3
	8pins	12
Screw connectors 5.08mm	2 poles	2
2x20 Female header extra height	For ***Raspberry Pi 4***, with 5.08mm additional spacer (taller version)	1

Screws list for the **wooden base version**

Screw (alternative screw)	Screw head	total	Thread
2x10mm (2.5x10mm)	flat	2	for wood
3,5x16mm (3,5x20mm)	conical	14	
3,5x30mm (3,5x25mm)	conical	5	
M2.5x6mm	flat	10	Metric
M3x10mm (M3x12mm)	flat	2	
M3x20mm (M3x25mm) + washers + nuts	flat	8	
M3x10mm (M3x8mm)	conical	12	
M3x10mm (M3x12mm)	conical	4	
M4x12mm (M4x16mm)	conical	16	
M4x20mm ((M4x25mm)	conical	1	
M4x30mm (M4x25mm)	conical	3	
M5x40mm (M5x35mm)	Conical	3	

Screws list for the **fully 3D printable version**

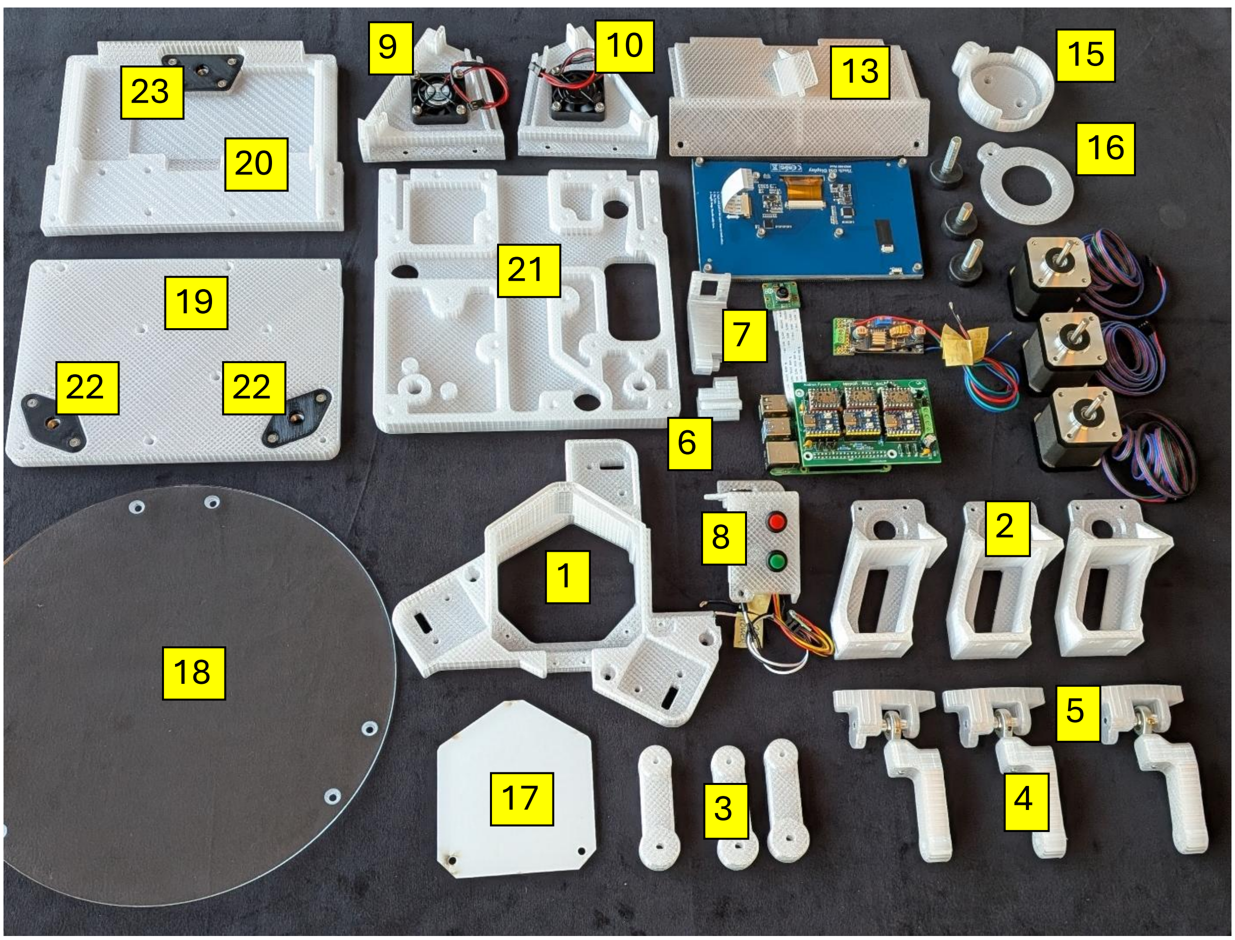
Screw (alternative screw)	Screw head	total	Thread
M2,5x6mm	Flat	10	Metric
M2,5x8mm (M2.5x10mm)	Flat	2	
M3x10mm (M3*12mm)	Flat	2	
M3x10mm (M3*8mm)	Conical	12	
M3x10mm (M3*12mm)	Conical	24	
M3x20mm (M3x25mm) + washers + nuts	Flat	8	
M4x12mm (up to max M4x20mm)	Conical	18	
M4x20mm (M4x16mm)	Conical	6	
M4x25mm (M4x20mm)	Conical	18	
M4x30mm (M4x25mm)	Conical	3	
M5x40mm (M5x35mm)	Conical	3	

Other parts, common for both the versions

Screw (alternative screw)	total	Notes
Nut (M5, or M6 or M8)	3	
Standoff screws: M2.5mm x 15mm x 6mm + M2.5 nut	4	Metric thread, brass or plastic
Cylindrical spacers: Øi=5, Øe=7~8mm, h=5mm	6	Spacers to keep the spheric joint fix, and in the middle of the ***Platform_support***'s slot These can be done by cutting an aluminium tube. Perhaps also 3D printed.

8. Parts' name

List of the (non-commercial) parts, to get some confidence



For both versions

Ref	Part name	Short description
1	Mirror_support	Supports the Mirror, Motor_supports, Camera_holder_low, Power_in
2	Motor_supports	Support the stepper motors
3	Arm_1	Arm connected to the motor shaft
4	Arm_2	Arm connected to the Arm_1
5	Platform support	Part in between Arm_2 and the Platform
6	Camera_holder_low	Supports the Camera_holder_top
7	Camera_holder_top	Supports the PiCamera
8	Power_in	Support the pushbuttons the power connector
9	Supp_Rpi_L	Left support for Display and electronic
10	Supp_Rpi_R	Right support for Display and electronic
11	PCB_top_cover	Top panel closing the electronic case (laser cut of 3D printed)
12	PCB_back_cover	Back panel closing the electronic case (laser cut of 3D printed)
13	PCB_combined_cover	3D printed cover closing the top and the back
14	Label_mbb or Label_neutral	Aesthetic part
15	Spirit_level_holder	Supports the spirit level and its cover
16	Spirit_level_cover	Retains the spirit level
17	Mirror	
18	Platform	

Specific parts for Fully 3D printable version

Ref	Part name	Short description
19	Base_bottom_R	The bottom base with the **R**ear feet
20	Base_bottom_F	The bottom base with the **F**ront foot
21	Base_top	The smaller base, that stays above the Base_bottom_R and Base_bottom_F
22	Rear_foot_support_Mx	Part holding the nut for the rear feet. Mx can be M5, M6 or M8
23	Front_foot_support_Mx	Part holding the nut for the front foot. Mx can be M5, M6 or M8

Specific parts for the wooden base version

Part name	Short description
Wooden_base_bottom	The largest wooden base, holding the feet
Wooden_base_top	The smaller wooden base, that stays above the Wooden_base_bottom
PCB_front_cover	Aesthetic part flashing the Display's bottom edge

9. Where to get files

All files are available on GitHub:
https://github.com/AndreaFavero71/MirrorBallBot

Version	Folder	Contents
Wooden base	/3d_models/wooden_base/stl	3D printable parts
	/3d_models/wooden_base/dxf	Laser cutting files (platform and mirror)
	/3d_models/wooden_base/stp	3D editable files
	/3d_models/wooden_base/pdf	2D drawing for the wooden base preparation
	/3d_models/wooden_base/complete_robot	Step file for the complete robot
Fully 3D printable	/3d_models/fully_3d_printable/stl	3D printable parts for fully 3D version
	/3d_models/fully_3d_printable/dxf	Laser cutting files
	/3d_models/fully_3d_printable/stp	Step file for the complete robot
	/3d_models/ fully_3d_printable /complete_robot	Step file for the complete robot
Both	/board	PCB Gerber files and schematic PDF
	/src	Python code for Raspberry Pi 4
	/rp2040	MicroPython code for RP2040-Zero boards
	/doc	Manuals

10. Build steps

This is the complete build checklist. Each step is detailed in the following chapters.

Step	**What to Do**
1	Choose your model (wooden base or fully 3D-printable)
2	3D print all parts (30% infill, no supports, 0.2mm layer, suggested PETG)
3	Laser cut the acrylic platform and mirror
4	Prepare the base (woodworking or 3D printing)
5	Assemble the MirrorBallBot PCB
6	Prepare the Power_in box with socket and buttons
7	**IMPORTANT:** Calibrate XL4015 to 5.1V BEFORE connecting to PCB
8	Flash all three RP2040-Zero boards with MicroPython v1.24
9	Set up Raspberry Pi 4 (flash OS, clone repo, run installer)
10	Run first tests (motors, fans, camera) – set TMC2209 Vref
11	Assemble the complete robot (follow assembly table)
12	Tune sensorless homing and platform levelling
13	Calibrate ball color (use AUTO calibration in GUI)
14	Tune PID settings (start with Kp=1.3, Ki=0, Kd=1.0)
15	Test predefined paths (square, circle)
16	Try freehand finger-following on the touchscreen
17	Enjoy your MirrorBallBot!

11. Choose your model

MirrorBallBot is available in two versions. Choose **before** you start 3D printing, as some parts differ between models.

Feature	**Wooden Base** (my preference)	**Fully 3D-Printed Base**
Construction	Two layers of 18mm plywood (CNC or manual router)	Fully 3D printed, split to fit 210×210mm beds
Robot weight	4.2 kg	3.1 kg
Pros	Heavier = dampens vibrations, classic aesthetic	No woodworking required, fully 3D printable
Cons	Requires woodworking tools or CNC	Doubling the 3D printing time
Filament needed	~550g (PETG)	~1100g (PETG)

My recommendation:

- **Wooden base** If you enjoy woodworking and have the tools
- **3D-printed base** If you don't have woodworking tools or want it fully 3D printed

Important: STL files are in separate folders (/wooden_base/stl and /fully_3d_printable/stl). Do **not** mix files between versions.

12. 3D printing

Material: PETG is recommended for its balance of strength and flexibility. It also doesn't degrade over time or from UV light.

Settings I used on my Ender 3 printer

Parameter	Setting
Layer height	0.2 mm
Infill	**30%** (essential for screw boss strength)
Supports	**None** (all parts designed to print support-free)
Bed temperature	82°C
Nozzle temperature	254°C
Vertical shells	2
Horizontal layers	3

Printing Time Reference (Ender3)

Version	Total Filament	Total Time
Wooden base	~500g	~46 hours
Fully 3D-printable	~1173g	~93 hours

Note: These times are for reference. Your printer may be faster or slower.

Files

Version	STL Files
Wooden base	/3d_models/wooden_base/stl
Fully 3D-printable	/3d_models/fully_3d_printable/stl
Version	STL Files

For editable files (STEP format), check the /step folders in the repository.

The complete robot, for both versions, is also available in the /complete_robot folders in the repository. This might be of interest for those who'd like deriving their own version.

13. Prepare the acrylic parts (Laser Cutting)

Materials

Part	Material	Thickness
Platform	Translucent acrylic/Plexiglass	3mm (balancing) or 6mm (bouncing)
Mirror	Acrylic mirror with self-adhesive backing	2.5mm

Cutting Specifications

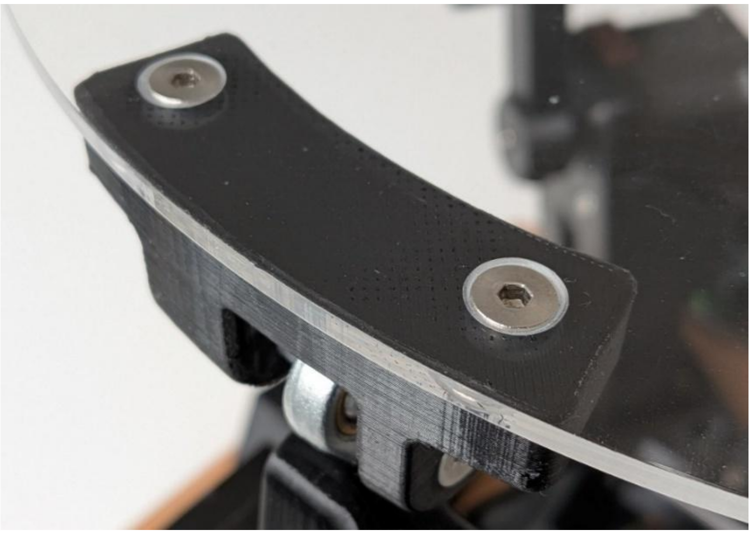
Part	Specification
Platform	Outer diameter 260mm, 6 holes Ø4.5mm (from DXF file)
Mirror	Cut to size (DXF file provided)

After Cutting

1. Chamfer the 6 holes in the platform with a chamfering drill bit
2. Remove protective film from both sides of the platform
3. Peel the mirror's protective film (after assembly)

Files

Version	DXF Location
Wooden base	/3d_models/wooden_base/dxf
Fully 3D-printable	/3d_models/fully_3d_printable/dxf



14. Prepare the Base

Wooden Base Version (skip if using 3D-printed base)

The wooden base consists of two 18mm plywood layers (top and bottom). This hides all cables between the layers.

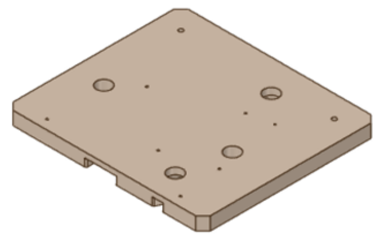
Files needed:

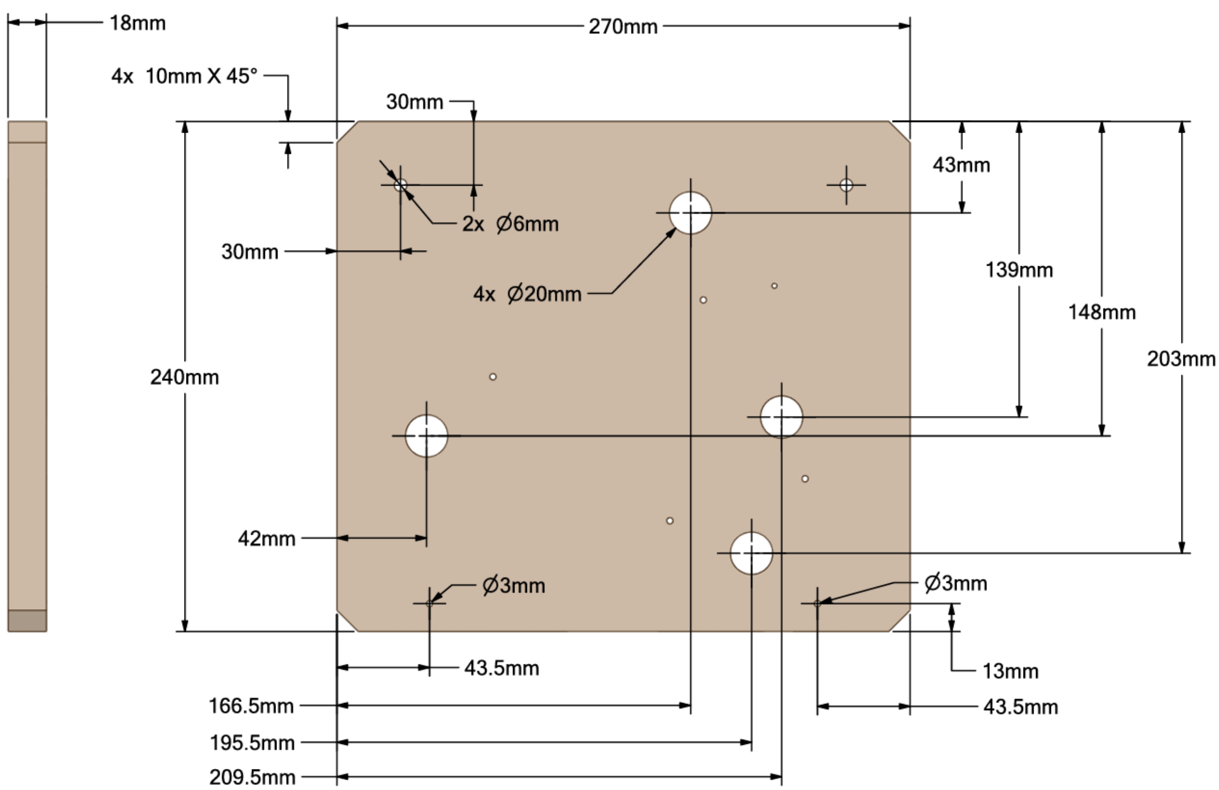
- 2D drawings (PDF): /3d_models/wooden_base/pdf
- STEP files for CNC: /3d_models/wooden_base/step

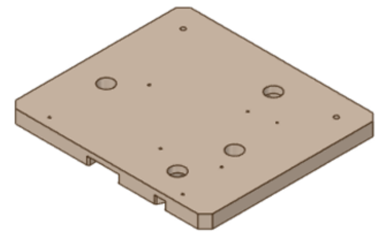
Notes: If using a manual router, clamp a straight wood piece as a guide.

Wooden_*base_top*

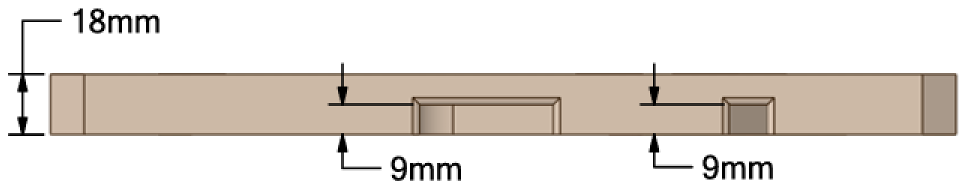
On below sketch the dimensions to realize the large holes (Ø20mm) for the motors' cables, those for the back feet (Ø depending on the feet rod diameter) and those to fix the ***Supp_Rpi_L*** and ***Supp_Rpi_R***:

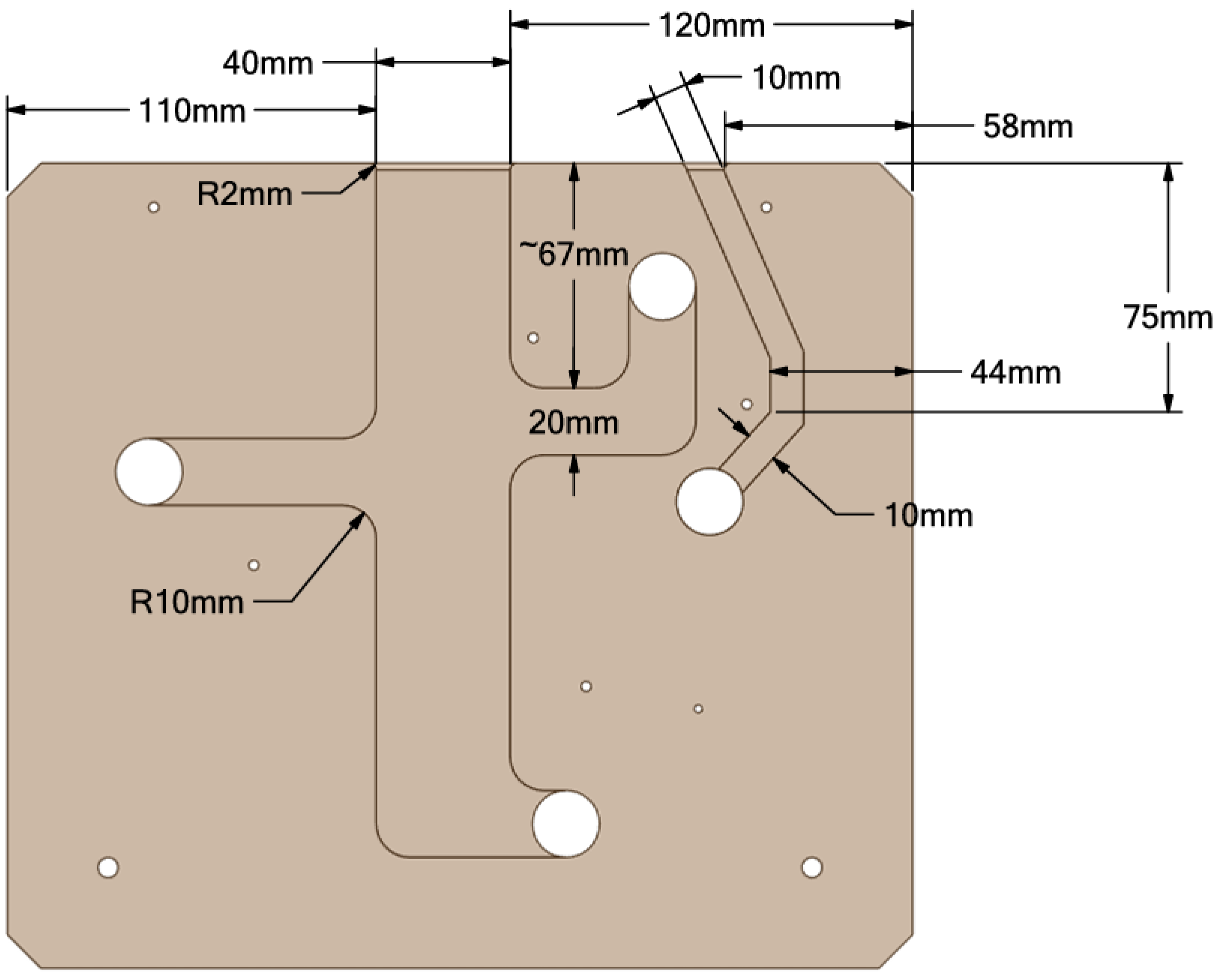




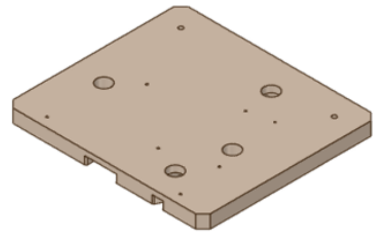


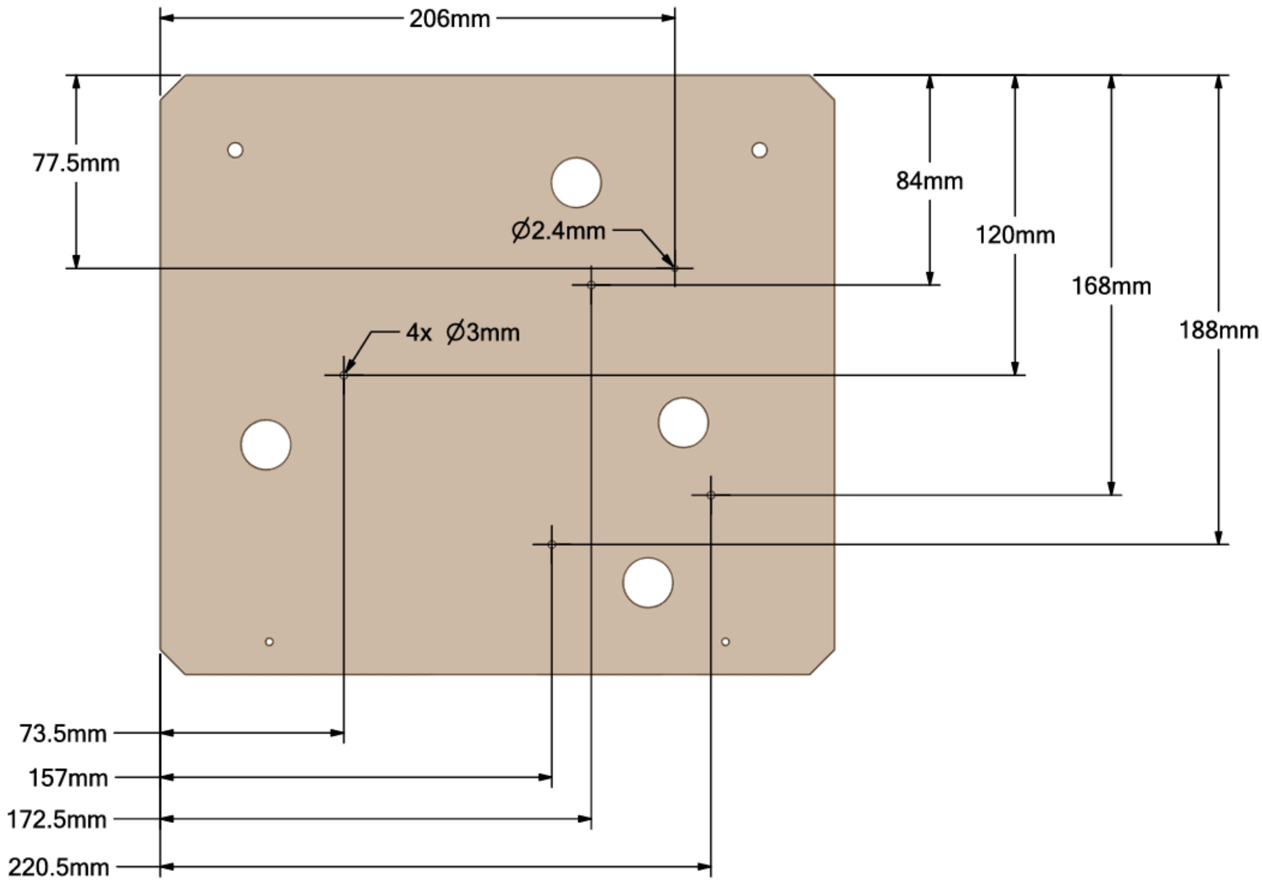
On below sketch the dimension to realize the grooves (depth ca 9mm) on the bottom side of the ***Wooden_base_top***.
I made the grooves via an electric hand router, with a cylindrical tool, by clamping a piece of straight wood as side guide. Suggested to remove only a couple of millimetres at the time.



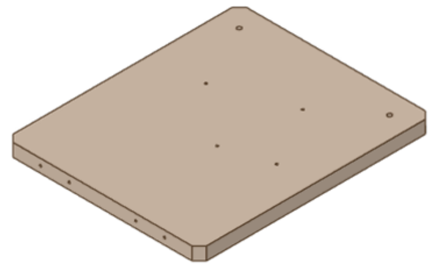


On below sketch the through holes Ø3 mm to fix the 3D printed “***Mirror_support***” and though hole Ø2.4 mm to fix the “***Power_in***”:



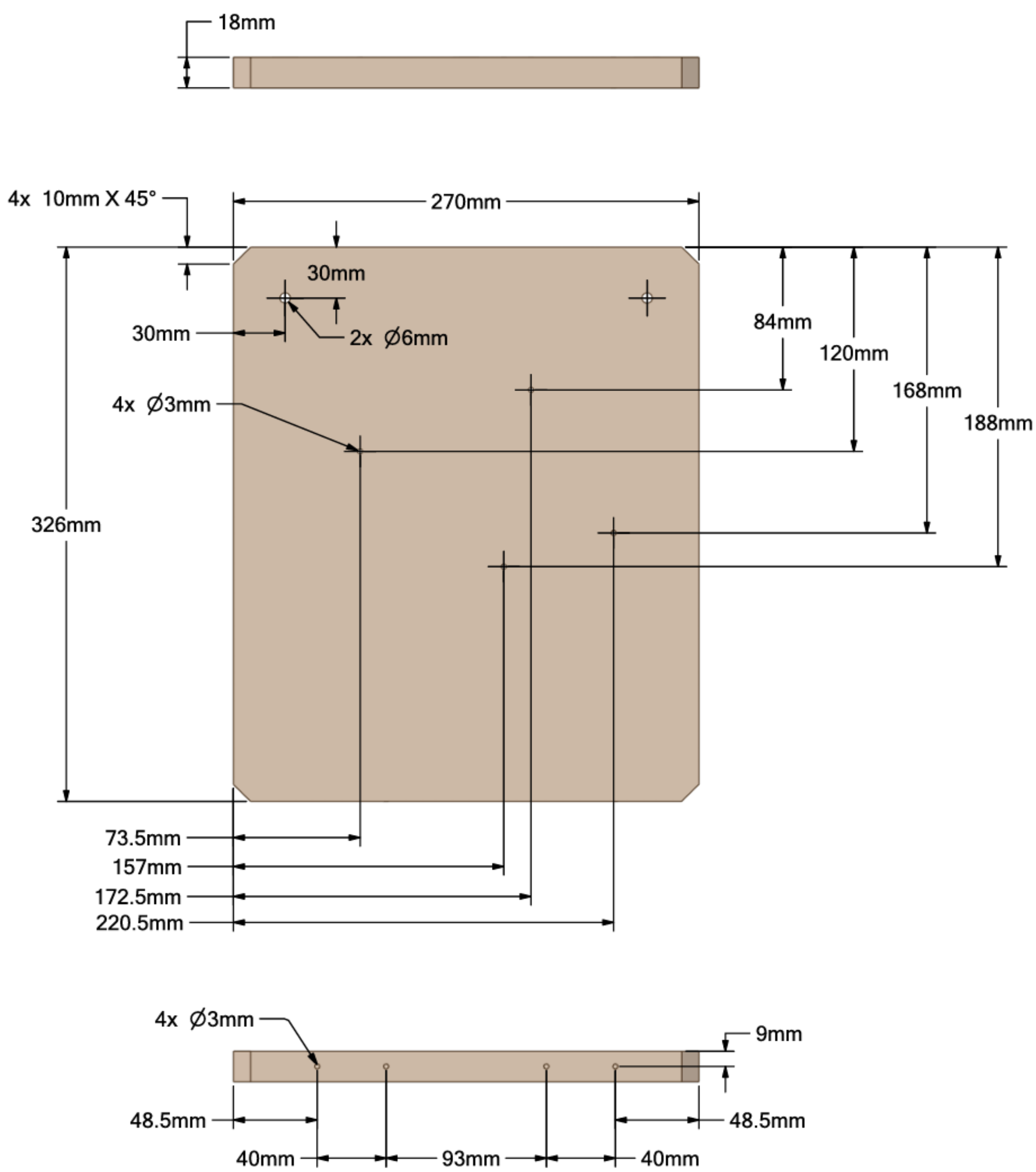


Wooden_base_bottom:



On below sketch the dimensions to realize the back feet (Ø depending on the feet rod diameter).
There are 4 holes as extension of the holes made on the ***Wooden_base_top***, to fix the ***Mirror_support***.
The holes extension enables the usage of longer screws, and it might serve as centring features between the ***Wooden_base_top*** and ***Wooden_base_bottom***.

The frontal holes to fix the ***Supp_Rpi_L***, ***Supp_Rpi_R*** and ***PCB_front_frame***.



After the individual wooden parts preparation:

1. Clamp both plywood layers together with good alignment
2. Drill 5 Ø3mm holes through both layers (outside the groove areas)
3. Use 3.5×30mm conical screws to join them (bottom-up)
4. Press nuts for feet into the bottom layer

Fully 3D-Printed Base Version (skip if using the wooden base version)

The base is split into three parts to fit a 210×210mm print bed.

Parts to print:

- Base_top.stl
- Base_bottom_R.stl (rear half)
- Base_bottom_F.stl (front half)

Foot supports:

- Rear_foot_support_Mx.stl (M5, M6, or M8 depending on your feet)
- Front_foot_support_Mx.stl

Important: The foot supports must be printed with the correct nut size (M5, M6, or M8) to match your adjustable feet.

15. Assemble the MirrorBallBot PCB

What the PCB Does

This is a passive board that connects all components together, eliminating the wiring jungle. It has ~250 solder pads—all through-hole components.

PCB Features

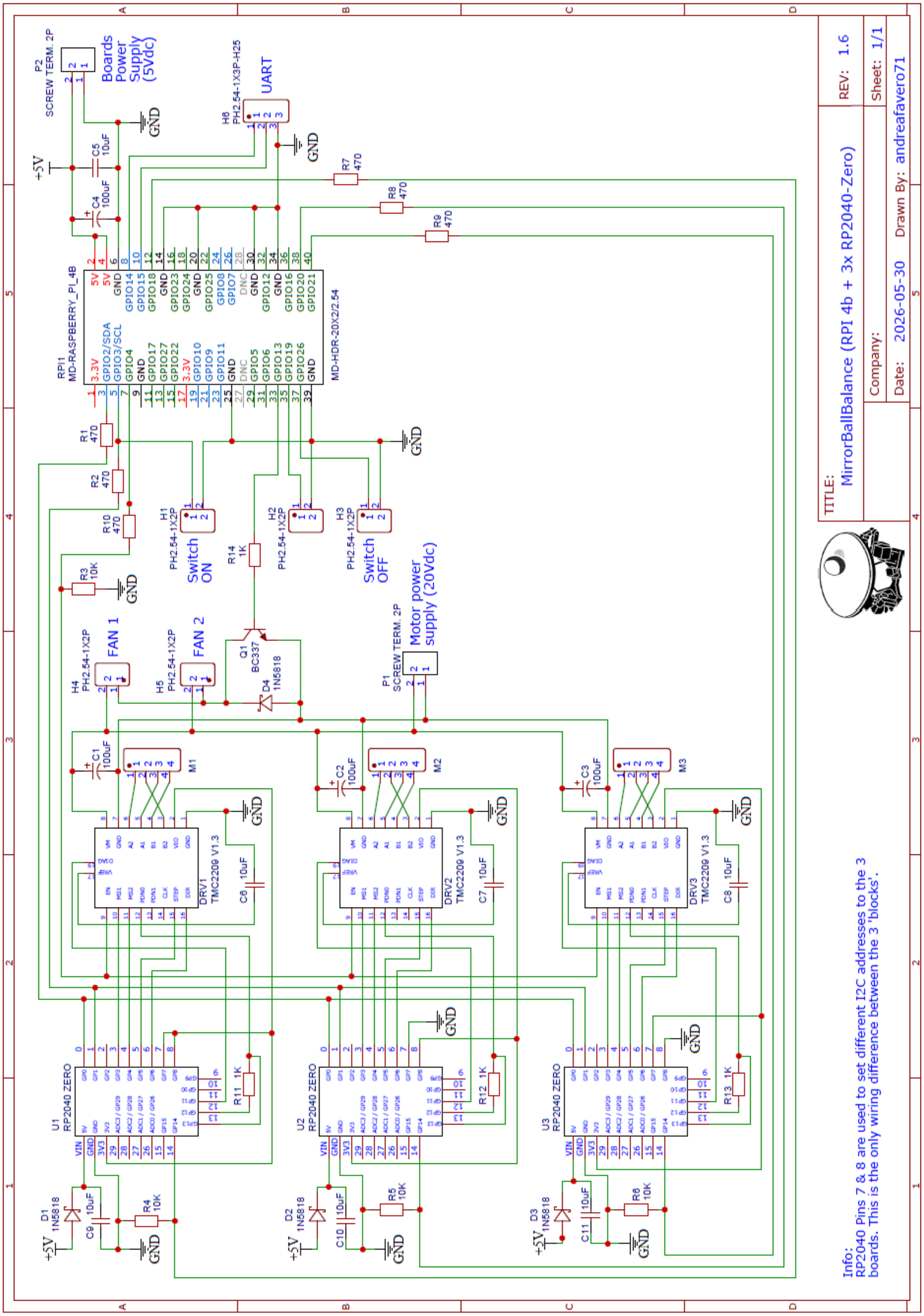
- Screw terminals for 20V (motors) and 5.1V (Raspberry Pi)
- Sets unique I2C addresses for each RP2040-Zero (via GPIO 7 and 8)
- Headers for stepper motors, fans, pushbuttons
- Connects RPi GPIO to RP2040-Zero boards
- Transistor for PWM fan control

Assembly Order

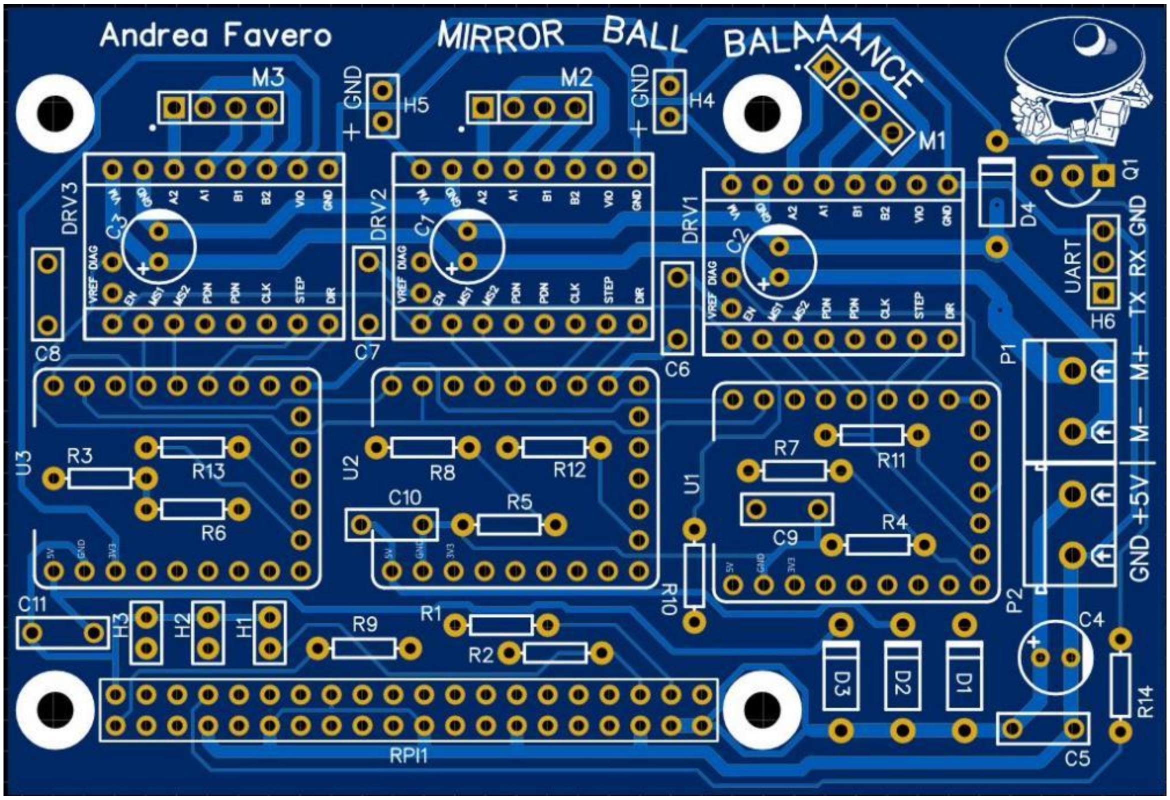
Step	Components	Notes
1	Resistors, diodes, ceramic capacitors	Start from center, move outward
2	Electrolytic capacitors (4×)	Lay on their side
3	Female headers	For RP2040-Zero (3 sides) and TMC2209
4	Male headers and screw terminals	Front side only
5	**2×20 female header (back side)**	**Must be taller version (+5.08mm spacer)**
6	Cut protruding pins from TMC2209	From the top side
7	Plug in TMC2209 drivers	Ensure correct orientation
8	Plug in RP2040-Zero boards	After programming (see Step 8)

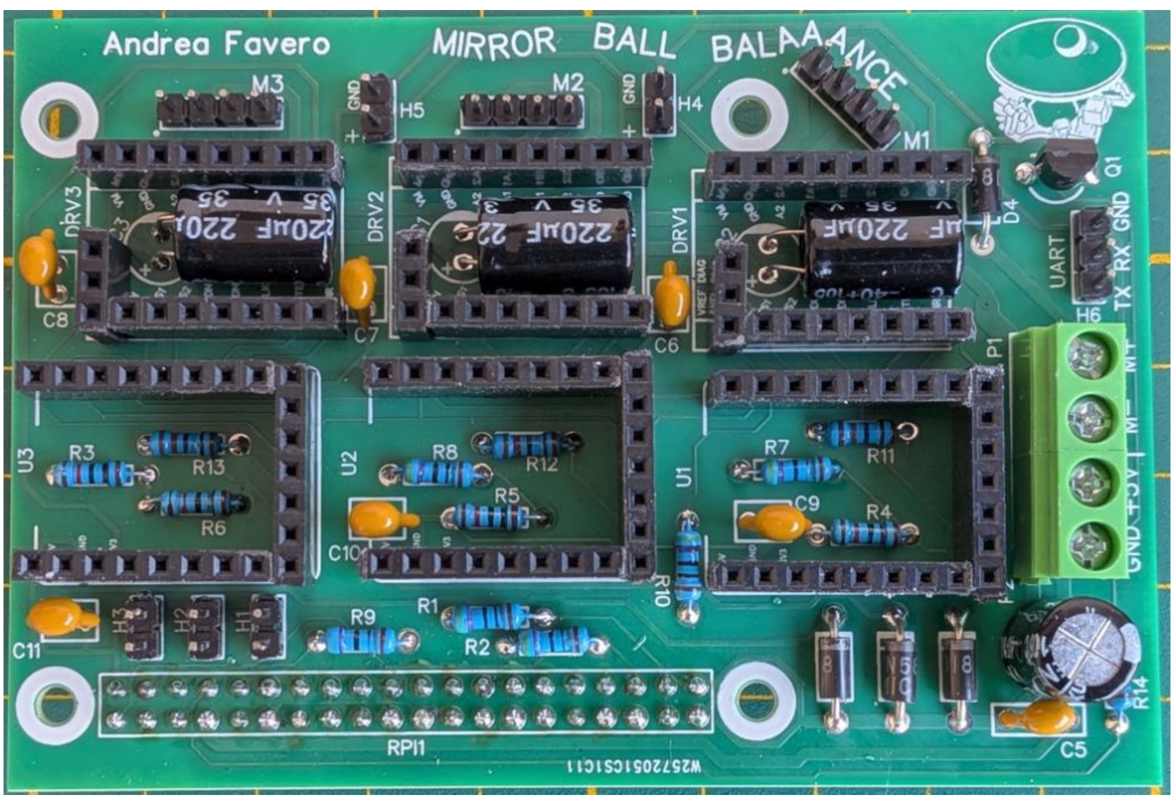
Warning: The 2×20 GPIO header for the Raspberry Pi must be the **taller version** (+5.08mm) to fit properly.

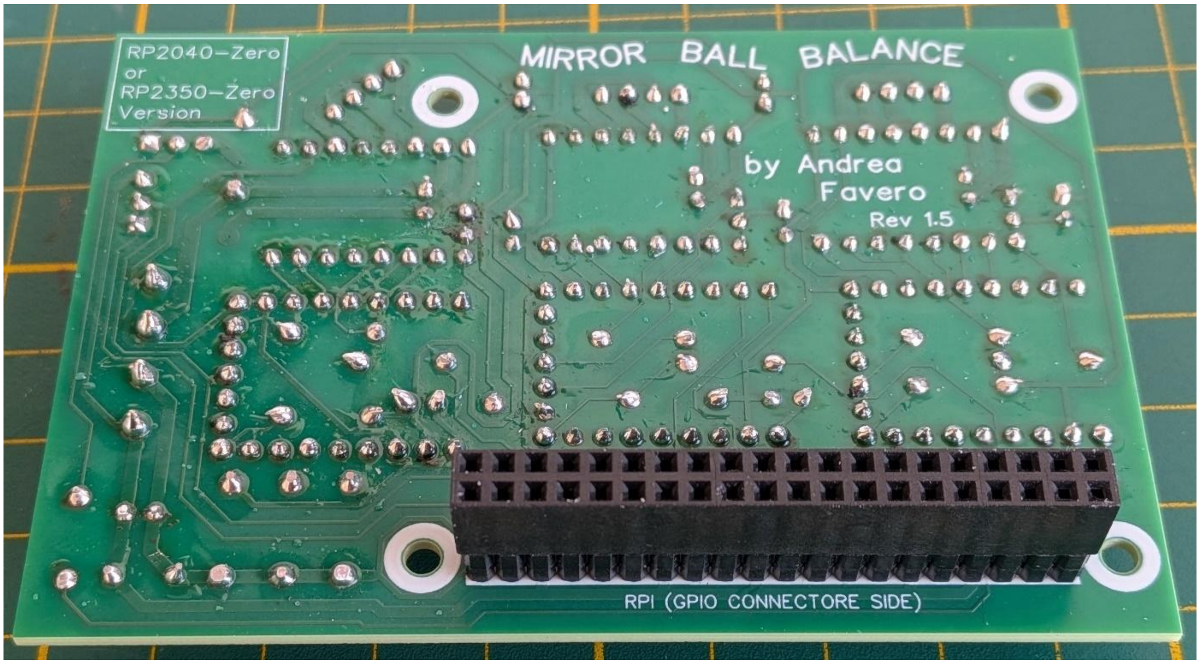
On next pages the schematic (also available as PDF in the ***/board*** folder), and some pictures of the board

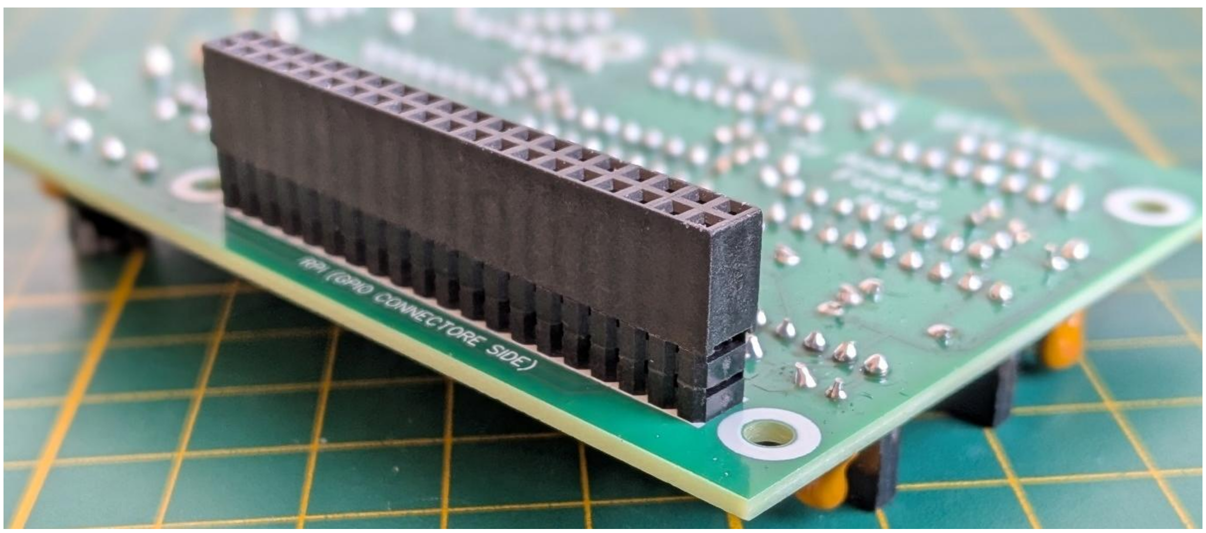

TITLE:
MirrorBallBalance (RPI 4b + 3x RP2040-Zero)
REV: 1.6
Company:
Sheet: 1/1
Date: 2026-05-30
Drawn By: andreafavero71
Info:
RP2040 Pins 7 & 8 are used to set different I2C addresses to the 3 boards. This is the only wiring difference between the 3 'blocks'.
Boards Power Supply (5Vdc)
UART
Switch ON
Switch OFF
FAN 1
FAN 2
Motor power supply (20Vdc)

PCB front side:






RP2040-Zero
or
RP2350-Zero
Version
MIRROR BALL BALANCE
by Andrea
Favero
Rev 1.5
RPI (GPIO CONNECTORE SIDE)


RPI (GPIO CONNECTORE SIDE)

16. Prepare the Power_in box

This 3D-printed box holds:

- Power socket (matching your 20V 5A power supply)
- Two pushbuttons (ON and OFF)

Pushbutton Function

- **OFF (red):** Safely shuts down Raspberry Pi OS
- **ON (green):** Powers on the Raspberry Pi (if previously shut down)

Wiring

1. Wires should be 30~35cm long.
2. Use a 3-wire cable for the pushbuttons.
3. Connect GND to one side of both pushbuttons.
4. Connect the other side to the remaining two wires.
5. Take note of which color is ON and which is OFF.
6. Extend the power cables, if not 30~35cm long

Customizing

The Power_in.stp file is provided so you can modify it to match your power supply connector.

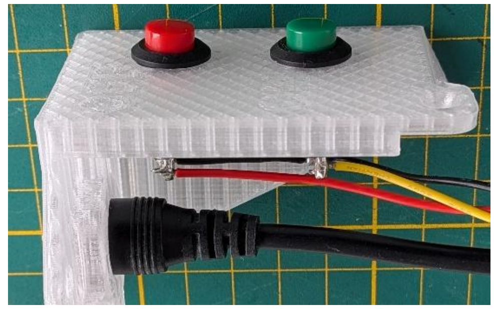
Tip: If using a laptop power supply like I did, you can use an adapter cable (7.4×5mm female to 5.5×2.5mm male). The socket simply presses into the 3D-printed case from the inside out.

The Socket is simply pressed in, from the inside toward the outside of the 3D printed case.

"Computer Adapter Cable Converter Cable 7.4x5mm Female to 5.5x2.5mm Male for Laptop"
https://tinyurl.com/bdz3cauv

You can edit the ***Power_in.stp*** file to redesign it to match your own power supply connector.

Red pushbutton is placed on the location with ***OFF*** engraved, the green one at the ***ON***:

Use a 3 wires cable, by connecting the GND wire to one side of both pushbuttons, and the other side to the other two cables. Take note of which colour is ON and which one is OFF.

17. Calibrate the XL4015 module

WARNING: DO NOT connect the Raspberry Pi to the PCB before adjusting the XL4015!

The XL4015 module ships with its output set to **maximum voltage** (~20V if using a 20V supply). This will **destroy** your Raspberry Pi in milliseconds if connected!

Calibration Procedure

1. **Do not connect** the XL4015 output to the PCB yet
2. Connect a **multimeter** to the XL4015 OUT+ and OUT-
3. Connect the XL4015 input to the Power_in power lines
4. Connect your 20V 5A power supply to the Power_in socket
5. The red LED on the XL4015 will light up
6. Using a precision screwdriver, adjust the potentiometer:
    - Turn **CCW** several turns until the LED dims (<3V)
    - Slowly turn **CW** to set the output to **5.1Vdc**
7. Disconnect the XL4015 input from the Power_in

Now it's safe to connect the XL4015 output to the PCB.

18. Set up RP2040-Zero boards

Each of the three boards controls one stepper motor.
All boards use the **same code:** The I2C address of each board is determined by the PCB position (via GPIO 7 and 8).

Download MicroPython v1.24 for Pimoroni Tiny 2040:
https://micropython.org/download/PIMORONI_TINY2040/

Process (repeat for each board)

1. **Connect** the RP2040-Zero via USB while holding the **BOOT** button
2. **Drag and drop** the .uf2 file to the RPI-RP2 drive
3. **Copy files** from the /rp2040 folder in the repository to the board:
    - main.py
    - rp2040_ball_balance.py
    - i2c_handler.py
    - i2c_responder.py
    - stepper_controller.py
    - shared_variables.py
    - shared_memory.py
    - rgb_led.py
    - TMC_2209_driver.py
    - TMC_2209_uart.py
4. **Verify** by pressing the RST button, the onboard RGB LED should flash alternating colors for ~5 seconds

Troubleshooting

If the board becomes unresponsive after booting (core1 takes over), you can:

- Press RST and immediately press STOP in Thonny, or
- Run the reset.py helper file

Updating Later
To update RP2040 code, you must remove each board from the PCB and connect via USB. No re-flashing of the firmware is needed—just copy the updated .py files.

19. Set up Raspberry Pi 4

⚠️ **WARNING** ⚠️ **DO NOT** connect the Raspberry Pi to the PCB before calibrating the XL4015 to 5.1V (see related chapter).

A. Flash the OS

1. Download **Raspberry Pi OS 32-bit Desktop** (Trixie recommended, Bookworm also works)
2. Use **Raspberry Pi Imager** to flash to your microSD card
3. Set during flashing:
    - Hostname (e.g., "mirrorballbot")
    - Wi-Fi credentials
    - Enable SSH

B. Update the OS

1. Insert the microSD into your Pi 4
2. Power the Pi with a standard 5.1V USB-C supply
3. Find the Pi's IP address (check your router or use an IP scanner)
4. Connect via SSH: ssh pi@<IP_ADDRESS> (or via Putty)
5. Run:

```
sudo apt update
sudo apt upgrade -y
sudo reboot
```

C. Clone the Repository

```
cd /home/pi
git clone https://github.com/AndreaFavero71/MirrorBallBot.git
```

D. Install Libraries

```
cd ~/MirrorBallBot/src
bash mbb_install.sh
```

This installs OpenCV, NumPy, smbus2, and other Python libraries. Takes 3-5 minutes with internet.

E. Enable VNC (optional)

Option 1: RealVNC (X11)

sudo raspi-config nonint do_wayland W1

sudo reboot

sudo systemctl enable vncserver-x11-serviced

sudo systemctl start vncserver-x11-serviced

Option 2: TigerVNC (Wayland)

sudo systemctl enable wayvnc

sudo systemctl start wayvnc

F. Enable GUI Auto-Start (optional, when robot is ready)

crontab -e

Uncomment the line ***@reboot bash -l/home/pi/MirrorBallBot/src/mbb_start.sh***

20. First tests

Run these tests **before** full assembly to catch issues early.

Before Powering On

1. **Visually inspect** the PCB for solder bridges
2. **Check for shorts** with an ohmmeter:
    - +5V to GND
    - +20V to GND
    - +5V to +20V
3. **Verify XL4015** is set to 5.1V
4. **Check polarity** on P1 (20V) and P2 (5.1V) terminals
5. **Ensure** RP2040-Zero and TMC2209 boards are fully inserted

Test Procedure

1. Power On

Connect the power supply. The Pi should boot and the display should show the desktop. If not, check troubleshooting in the full manual. The RP2040-Zero onboard RGB LED should flash alternating colors for ~5 seconds

2. Camera Test

cd ~/MirrorBallBot/src

python3 mbb_camera.py

- You should see the camera image on screen
- FPS should be above 80 (optimized >110 FPS in GUI mode)
- Press ***a*** (the graphic window should be active) for auto HSV calibration (place the ball in the centre first)
- Press ***q*** (the graphic window should be active) to quit

3. Fans Test

python3 mbb_fans_test.py

- Fans should run at various speeds
- Test cycles automatically and can be interrupted with CTRL+C
- Ensure correct airflow direction (left fan = intake, right fan = exhaust)

4. Motor Test

IMPORTANT: Motors must **not** have arms attached for this test!

python3 mbb_motors_test.py

- Motors should rotate slowly CCW, then faster CW
- If all motors rotate backwards, reverse the reverse parameter in ***mbb_settings.json***
- If only 1-2 motors are reversed, rotate the connector 180° on the PCB

Set Vref on TMC2209:

1. Power down the system
2. Measure voltage between VREF pin and GND
3. Adjust the tiny potentiometer to **1.6V** (for 2A motors at 80% current)
4. Formula: Vref = Imax × 80% (e.g., 2.0A × 0.8 = 1.6V)

Note: If your TMC2209 has different shunt resistors (110mΩ is typical), the Vref may vary. Check the formula in the full manual.

Test Complete?

If all tests pass, you're ready for **full assembly**.

21. Full assembly

This table summarizes the assembly steps. For detailed instructions and pictures, refer to the **Full Reference Manual**.

Before you start: Ensure all first tests passed.

Screw tip:

- In case screw oppose high torque, rub them on candle wax to reduce friction.
- Do not over-tighten.

Suggested steps and assembly order

(*) when specific for Fully 3D printed version, (**) when specific for the Wooden base ver.

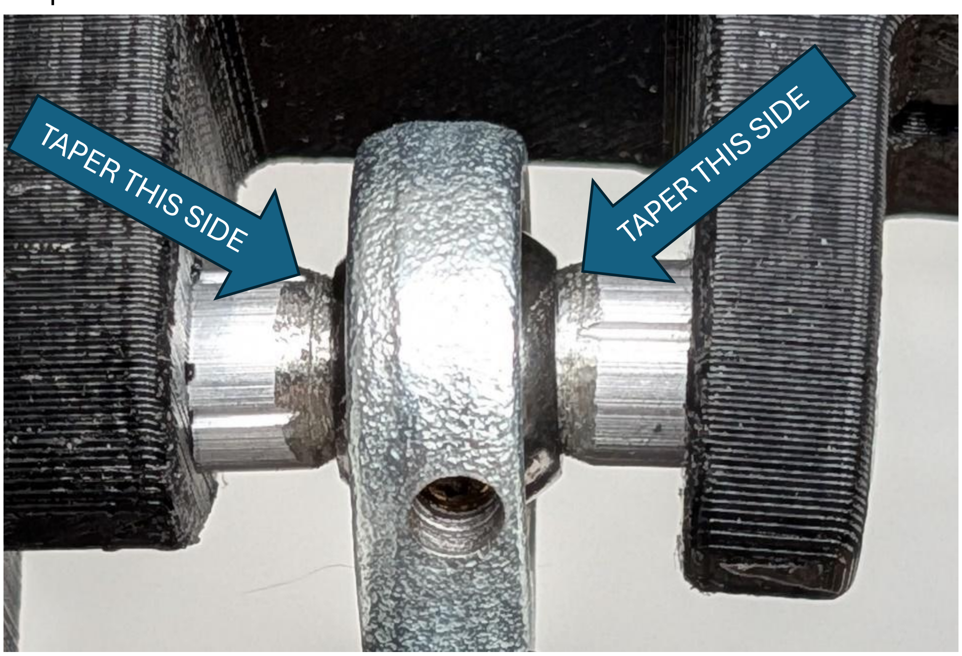
Step	Part	Part	Action / screw (alternative screw) and head type
1	Mirror	***Mirror_support***	2x M3x10mm (M3x12mm) flat
2	***Mirror_support***	***(*) Base_top***	4x M4x20mm conical
		(**) Wooden_base_top	4x 3,5x16mm (3,5x20mm) flat
3	***Motor_support***	***Mirror_support***	6x M4x12mm conical
4	Stepper motor	***Motor_support***	12x M3x10mm (M3x8mm) conical
5	Motors cables	***Stepper motors***	Mark the cables (A, B, C), pass them through the top base. Keep 25cm of cables protruding in front, block the rest within the grooves with tape.
6	***Power_in***	***Mirror_support***	3x M3x10mm (M3*12mm) conical
		(*) Base_top	1x M3x10mm (M3*12mm) conical
		(**) Wooden_base_top	1x 3,5x16mm (3,5x20mm) conical
7 (*)	***Front_foot_support***	***Base_bottom_F***	Press the (M5, M6 or M8) nut into the slot. Fix the supports to the bases: 8x M3x10mm (M3*12mm) conical
	Rear_foot_support	***Base_bottom_R***	

Step	Part	Part	Action / screw (alternative screw) and head type
8 (*)	***Base_bottom_R***	***Base_top***	17x M4x25mm (M4x20mm) conical
	Base_bottom_F		
8 (**)	***Wooden_base_bottom***	***Wooden_base_top***	5x 3,5x30mm (3,5x25mm) conical
9	bottom base (wood or 3D)	feet	Screw in the feet
10	***Camera_holder_low***	***Camera_holder_top***	1x M4x20mm (M4x35mm) conical
11	***PiCamera***	***Camera_holder_top***	Enlarge holes at PiCamera to Ø2.5mm. Screw: 2x M2.5x6mm flat
12	***Camera_holder_low***	***Mirror_support***	2x M3x10mm (M3x12mm) conical
13	***Spirit_level_holder***	***(*) Base_top***	2x M4x 12mm conical
		(**) Wooden_base_top	2x 3.5x16mm (3,5x20mm) conical
14	***Spirit_level_cover***	***Spirit_level_holder***	1x M4x12mm conical (or flat)
15	***Arm_1***	***Stepper motors***	Press the arm onto the D motor's axis. Keep the arms ~4mm from the motor. 3x M4x12mm conical
16	***Arm_2***	Spheric joints	Screw the M5 spheric joints to the Arm_2 (check if left-hand or right-handed thread)
17	Spheric joints	***Platform_support***	3x M5x30mm (M5x25mm) 6x Spacers Ø5 x Ø7~8 x 5mm screw with spacers at the spheric joints sides (see picture)
18	***Arm_2***	Arm1	Rub a candle (wax) on the Arms flat surfaces. Screw 3x M4x30mm (M4x25mm) conical, do not tighten!
19	***Platform_support***	***Platform***	Chamfer the six Platform's holes; Screw 6x M4x12mm conical (M4x16mm if the Platform is 6mm thick)

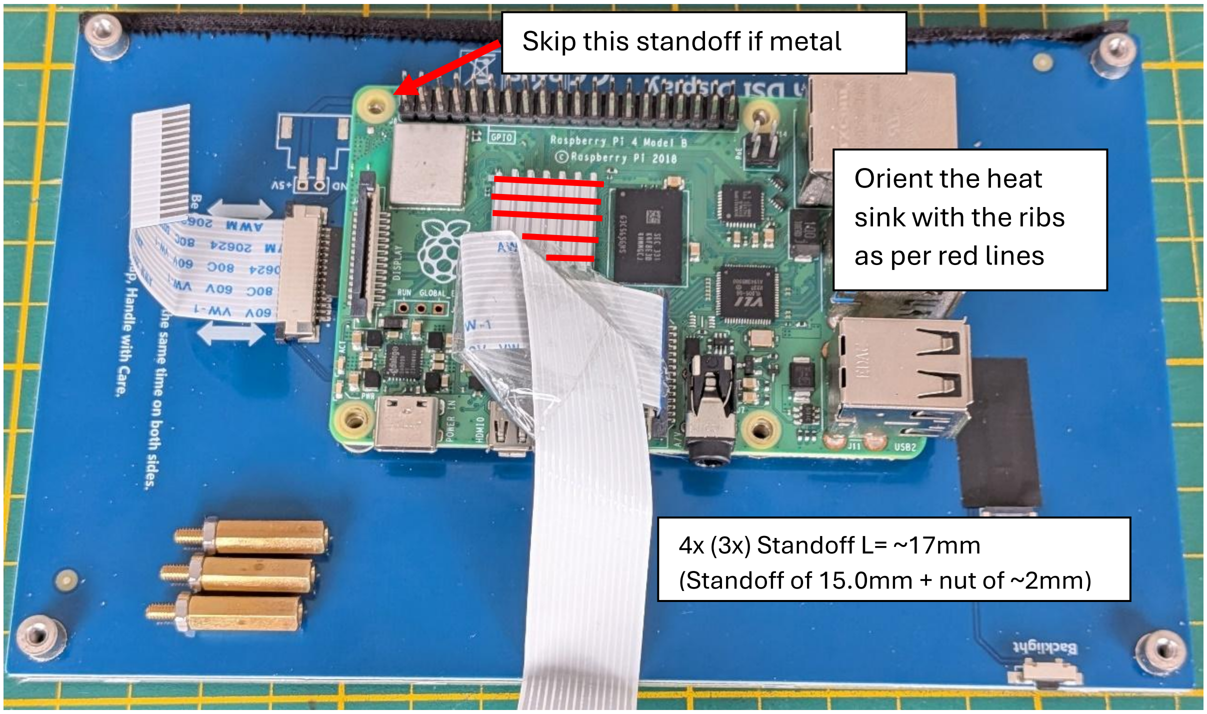
Step	Part	Part	Action / notes
20	Fan + grids Fan's label toward the inside (electronic).	***Supp_Rpi_L***	Grids on both sides. 4x M3x20mm (m3x25mm) flat + washers + nuts
21	Fan + grids Fan's label toward the outside	***Supp_Rpi_R***	Grids on both sides. 4x M3x20mm (m3x25mm) flat + washers + nuts
22	***XL4015*** step down module	***Power_in***	Connect the IN+ and IN- to the power lines (polarity matters!) Adjust the PWR_OUT to 5.1Vdc, if not done yet
23	***XL4015*** step down	***(*) Base_bottom_F***	2x M2,5x8mm (M2,5x10mm) flat Do not tighten
		(**) Wooden_base_bottom	2x M2x10mm flat, do not tighten
24	***Raspberry Pi 4***	Camera ribbon's cable	Add heatsink to the Rpi 4. Connect the ribbon-cable to the Display Connect the Camera's ribbon to the Rpi, fold it to stay flat and protect it with some tape against the MBB board protruding pins.
25	Standoff screws: M2.5mm x 15mm x 6mm + M2.5 nut	***Raspberry Pi 4***	If metal Standoff, use only 3 (see pictures), to limit negative impact to the Rpi Wi-Fi antenna
26	***Raspberry Pi 4***	DSI-display	Connect the Ribbon-cable to the Display's DSI connector
27	***Supp_Rpi_L*** ***Supp_Rpi_R***	DSI-display	4x M2.5x 6mm flat
28	***MirrorBallBot*** PCB (inclusive of ***RP2040-Zero*** and TMC2209)	***Raspberry Pi 4***	Connect the input power cables to the MBB board. Connect the MBB board to the Rpi GPIO, by centring on the screwing holes. Screw the MBB board: 4x (3x) M2.5x6mm flat
29	Fans	***MirrorBallBot*** PCB	Connect the fan's (Polarity matters!)

Step	Part	Part	Action / notes
30	Cables from ON /OFF pushbutton	***MirrorBallBot*** PCB	Prepare a single connector, and connect it to the MBB board
31	TMC2209	Heatsink	Cut the protruding pins. Add the heatsink
32	Motor's cable	***MirrorBallBot*** PCB	Connect the motors' cables
33	***Supp_Rpi_L*** ***Supp_Rpi_R***	***(*) Base_bottom_F***	6x M3x10mm (M3*12mm) conical
		(**) Wooden_base_bottom	4x 3.5x16mm (3,5x20mm) conical
34	Camera's ribbon	***PiCamera***	Connect the ribbon (contacts toward the camera's pcb)
35	***PCB_combined_cover*** or ***PCB_top_cover*** + ***PCB_back_cover***	***Supp_Rpi_L*** ***Supp_Rpi_R***	2x M3x10mm (M3*12mm) conical
36	***Label_mbb*** or ***Label_neutral*** (personalized…)	***(*) Base_bottom_F***	2x M3x10mm (M3x12mm) conical
	Label_mbb or ***Label_neutral*** (personalized…) + ***PCB_frame_front***	**(**) Wooden_base_bottom**	2x 3.5x16mm (3,5x20mm) conical

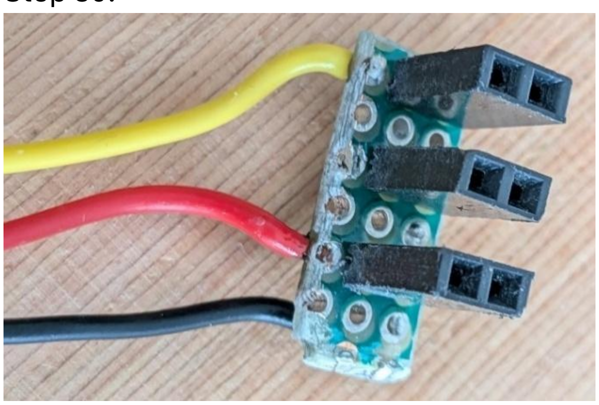
Step 17:

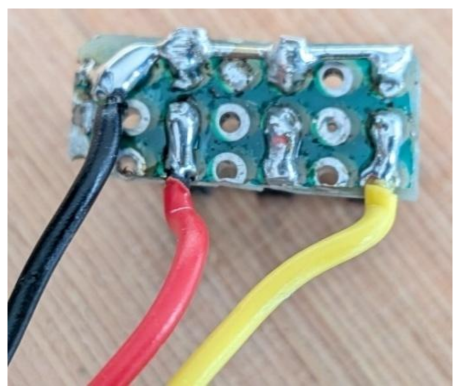


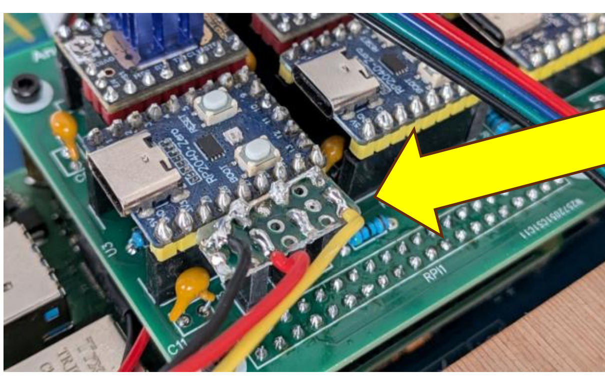
Step 24:



Step 30:



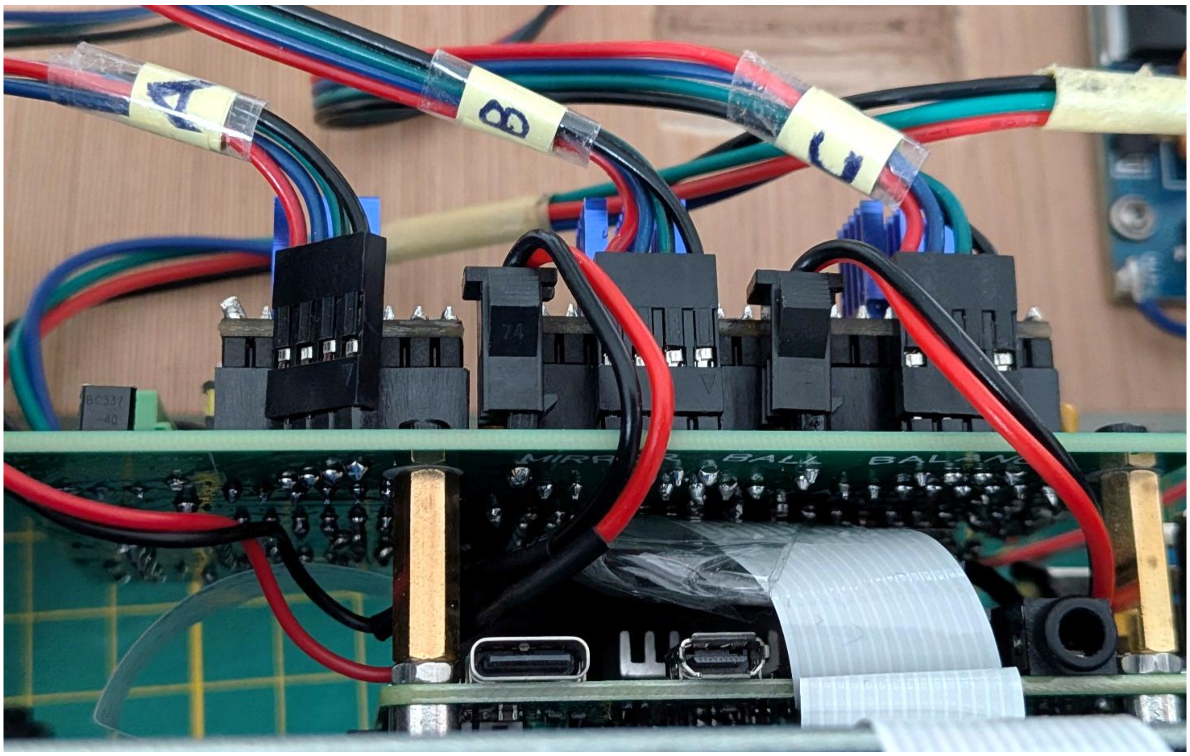




Step 32 (**Connect the motors cables)**
The orientation (cable's colour order) should be the same for the 3 connectors
If the three motors rotate on the opposite direction (see "First tests" chapter), rotate the 3 connectors, or change the setting in the ***mbb_settings.json*** file.

In case only one or two motors rotate on the opposite direction, then only those connectors must be rotated.

If one or more motors don't rotate smoothly, check the troubleshooting chapter.



Step 36

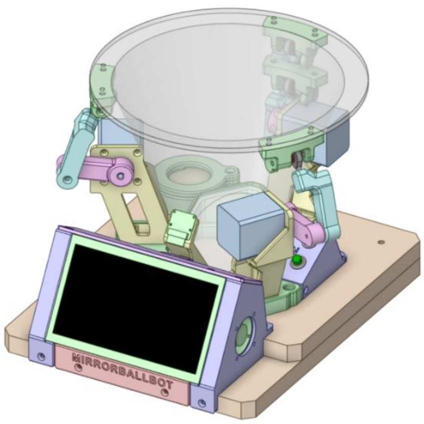
The ***PCB_front_frame*** is meant for aesthetic reasons, to get the Display and the ***Wooden_base_bottom*** flashed together.

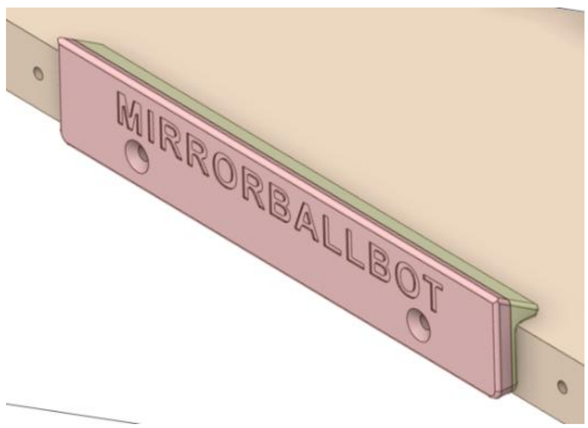
The ***Label_mbb*** (or ***Label_neutral*** personalized by you) is also meant for aesthetic reasons.

Wooden_base version:

In this version the ***Label_mbb***, or the ***Label_neutral*** (personalized by you) is applied over the ***PCB_front_cover***, used to "flash" the Display bottom edge.

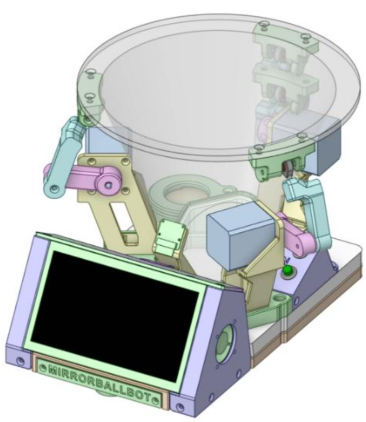
These two parts are kept separate to make printing easier and to achieve a better recessed text effect

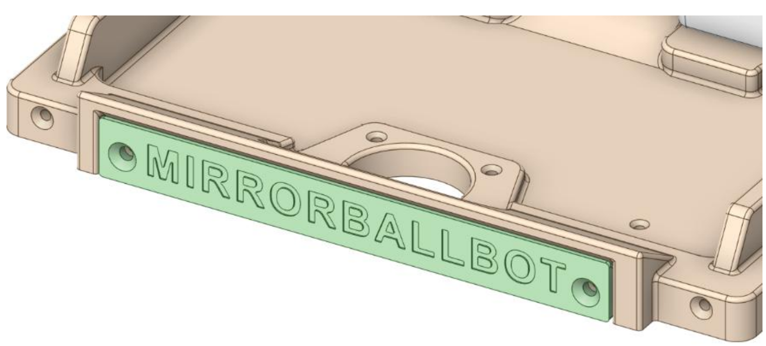




Fully 3D printable version

The ***Label_mbb***, or the ***Label_neutral*** (personalized by you) is fix directly to the ***Base_bottom_F***:





Congratulations

your ***MirrorBallBot*** is fully assembled:

Fully 3D printable version
(smaller base)

22. Calibration (brief)

This is the minimal calibration needed to get the robot working. For detailed tuning, refer to the full manual for more info.

Prerequisites:

- All first tests passed
- Robot fully assembled
- Power connected and RPi booted

Start the GUI

cd ~/MirrorBallBot/src
python3 mbb_gui.py

The GUI has three main tabs: **MAIN**, **PATHS**, and **CALIBRATION**.

Sensorless Homing

This tells the robot where the mechanical stops are.

1. Go to **CALIBRATION → ROBOT → HOMING**
2. Press **"Home All Motors"**
3. Each motor will rotate until it hits its stop
4. Watch the platform, all three supports should pull against the motors
5. Motor bounces back (skips steps, hits too hard), **increase** SG Factor (less sensitive)
6. Motor stops too early (false StallGuard trigger), **decrease** SG Factor (more sensitive)

Goal: All three motors stop consistently at the same position every time.

Platform Levelling

1. Place the spirit level on the platform
2. Go to **CALIBRATION → ROBOT → PLATFORM**
3. Adjust the three "Platform levelling (steps)" values
4. Press **"Level Platform"** to re-home and re-level
5. Repeat until the platform is level

Tip: Small adjustments (1-2 steps) make a noticeable difference.

Ball color calibration

1. Place the ball at the centre of the platform
2. Go to **CALIBRATION → BALL**
3. Ensure the "Ball Diameter (mm)" roughly matches your ball (default 30mm)
4. Press **"AUTO"**
5. Wait 10-20 seconds for automatic calibration
6. A green contour should appear around the ball

If AUTO fails:

- Ensure the ball is a solid color (not white, no patterns)
- Ensure the ball contrasts with the background
- Try the **"MANUAL"** calibration with the sliders
- Press **SAVE** to store the settings

PID Tuning (Basic)

1. Go to **CALIBRATION → ROBOT → PID**
2. Start with these values:
    - **Kp = 1.3**
    - **Ki = 0.0**
    - **Kd = 1.0**
3. Go to the **MAIN** page and enable **"Enable AUTO-Balance"**
4. The ball should move toward the centre

Fine-tuning:

- If the ball **oscillates** (moves back and forth): **increase Kd** (0.1-0.2 at a time)
- If the ball **drifts** slowly: **increase Ki** (start with 0.05)
- If the ball is **slow to respond**: **increase Kp** (0.1 at a time)

Note: Ki is optional. If the platform is well levelled, Ki=0 works perfectly.

23. Using the robot

Starting the GUI

cd ~/MirrorBallBot/src

python3 mbb_gui.py

or double tap on the MirrorBallBot icon on the Desktop

Main Controls

Control	Function
Enable AUTO-Balance	Activates ball balancing
Home Motors	Performs sensorless homing
Balance Plat.	Moves platform to level position
Emergency Stop	Cuts power to motors (red/green button)
Target Position	Move ball to specific X/Y coordinates (mm)
Set (0,0)	Returns ball to centre

Running a Path

1. Tap the **PATHS** tab
2. Select a path: **SQUARE**, **CIRCLE**, **INFINITY**, **TRIANGLE**, or **LINE**
3. Adjust size and repeats as desired
4. Tap **START** to begin

Finger Following (Free Path)

1. Go to **PATHS → FREE**
2. Touch the screen anywhere on the camera image
3. The ball will move to your finger position
4. Drag your finger to move the ball in real-time

Record a custom path:

1. Press **START REC**
2. Move your finger to create a path
3. Press **STOP REC**
4. Press **PLAY PATH** to replay

Notes:

- To exit fullscreen and access the desktop, tap the ▼ button at the bottom of the GUI.
- To change the Display's backlight, press the PCB_top_cover at the recess (10 levels).

24. Next steps

Congratulations! You've built and calibrated your MirrorBallBot.

Where to go from here:

Resource	What You'll Find
Reference Manual	Detailed explanations, theory, design process, troubleshooting, Q&A
GitHub Repository	Latest code, STL files, PCB files, and updates **Provide feedback** 😊 **If you enjoyed the project, consider buying me a coffee** 😊

If something isn't working:
The Quick Start Guide intentionally omits detailed troubleshooting. Refer to the **Reference Manual.**

If you want to understand the robot better:
The **Reference Manual (Section 4)** covers:

- Inspiration and design process
- The custom I2C protocol
- PID controller theory
- Software architecture

Feedback:
If you find errors or need better explanations, please let me know:

- **GitHub Issues:** https://github.com/AndreaFavero71/MirrorBallBot/issues
- **Email:** andrea.favero71@gmail.com

Enjoy your MirrorBallBot!

25. Revisions

Rev	Date	Notes
0	09/09/2026	First release